

Glitch Hunter — Tech Stack

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What this is

Glitch Hunter is a PPO reinforcement-learning agent that plays a Super Mario Bros clone and is watched live through a web dashboard; a second pipeline turns anything it catches breaking the game's own rules into an evidence bundle (screenshot, GIF, Markdown/PDF report). The table below lists every technology in the project and exactly which part of it uses that technology.

Stack by objective

Each objective built on the one before it: nothing was replaced, only added.

Objective	What it produced	Technologies it added
1 — Autonomous exploration agent	the 6M brain (46.8% level completion)	Python 3.12, Pygame, Gymnasium, PyTorch, Stable-Baselines3 PPO (CnnPolicy), NumPy, cloudpickle, Flask + Flask-SocketIO dashboard, OpenCV frame stream, HTML/CSS/JS, pytest, Git/GitHub
2 — QA / glitch-hunting agent	the 16M QA brain (84.25% coverage, 56.6% completion)	shared-memory coverage bitmap (NumPy + multiprocessing), reachability mask (.npz), QA reward and lifecycle, anchor-KL consolidation callback, 500-episode evaluation (multiprocessing Pool), SHA-256 artifact manifest, ruff, mypy, pytest-cov, GitHub Actions, uv
3 — Automated bug reporting	incidents with evidence, GIF/MD/PDF reports, replay; 6 benchmark bugs, all detected; a clean run that reaches the castle gets a "No Bugs Found" run report	incident pipeline and store, Pillow (GIF), fpdf2 (PDF), subprocess replay, zip bundles, REST API + Socket.IO events, game variants (mario_clean / mario_bugged), collision and jump-physics detectors, level-design reference file

Language, runtime and environment

Technology	Version	What it is	Where it is used	Objectives
Python	3.12 (exactly)	the programming language	the whole codebase: game wrapper, training, dashboard, reporting, tools, tests	1, 2, 3
venv (<code>venv_gpu/</code>)	—	isolated Python environment	holds every dependency; <code>run_dashboard.bat</code> and all commands use <code>venv_gpu\Scripts\python.exe</code>	1, 2, 3

Technology	Version	What it is	Where it is used	Objectives
pip + <code>requirements.txt</code>	—	package installer	the 2-step manual install (torch first, then <code>requirements.txt</code>)	1, 2, 3
uv + <code>uv.lock</code>	—	fast, locked installer	one-command install (<code>uv sync</code>) that encodes the torch/CUDA rules	2, 3
NVIDIA CUDA 12.6	cu126 build	GPU computing	fast PPO training on the GPU (CPU-only also works, just slower)	1, 2
Windows / Linux	Windows 11, Ubuntu (CI)	operating systems	development and dashboard on Windows; CI tests on both	1, 2, 3
Python standard library	3.12	built-in modules	<code>multiprocessing</code> (training workers, shared memory, parallel evaluation), <code>threading / queue</code> (dashboard game thread, report worker), <code>subprocess</code> (incident replay), <code>hashlib</code> (SHA-256), <code>json</code> , <code>zipfile</code> , <code>difflib</code> (bug-diff pin), <code>argparse</code> (tools), <code>ctypes</code> (Windows window control, power settings)	1, 2, 3

Game engine and simulation

Technology	Version	What it is	Where it is used	Objectives
Pygame	2.6.1	2D game library (on SDL 2.28)	runs the Super Mario Bros clone: drawing, sprites, collisions, the game window	1, 2, 3
Mario clone (<code>mario_clean/</code>)	vendored, pinned by SHA-256	the game itself (Level 1-1)	the untouched baseline every brain is trained and measured on	1, 2, 3
<code>mario_bugged/</code>	copy of the clone	the game with 6 declared benchmark bugs	bug-detection testing; picked from the dashboard's list	3
Gymnasium	1.3.0	standard RL environment API	<code>custom_mario_env.py</code> turns the game into an environment: 10 discrete actions, <code>step()</code> / <code>reset()</code> , info dict	1, 2, 3
Gymnasium wrappers	1.3.0	observation pipeline	frame skip 4, grayscale, resize to 84×84, stack of 4 frames (<code>wrap_observation</code>); <code>TimeLimit</code> in evaluation	1, 2, 3
SDL dummy video driver	—	headless rendering	training workers, evaluation, tests and incident replay run with no window	1, 2, 3

Technology	Version	What it is	Where it is used	Objectives
Windows user32 via ctypes	—	native window control	<code>game_window.py</code> : centre, show, hide and minimise the dashboard's game window to the taskbar; <code>desktop.py</code> : centre the terminal and browser windows, and keep full speed on battery (power mode, process throttling)	1, 3

Reinforcement learning and machine learning

Technology	Version	What it is	Where it is used	Objectives
PyTorch	2.14.0 (+cu126)	deep- learning framework	the neural network inside the agent; runs on the GPU in training, CPU in evaluation	1, 2, 3
Stable-Baselines3 — PPO	2.9.0	RL algorithm library	trains the agent (<code>train_agent.py</code>) and loads the brain for the dashboard and evaluation	1, 2, 3
CnnPolicy	SB3	convolutional policy + value network	reads the 84×84×4 frames and chooses one of 10 actions	1, 2, 3
SubprocVecEnv / DummyVecEnv	SB3	vectorised environments	8 parallel game workers in training; a single env in the dashboard	1, 2
SB3 callbacks (custom)	project	hooks into training	checkpoints with coverage, anchor-KL consolidation, stagnation watch, value warm-up (<code>training/callbacks.py</code>)	1, 2
Anchor-KL consolidation	project	keeps a skill from being forgotten	after each update, pulls the policy back toward a frozen healthy brain if KL > 0.08	2
cloudpickle	3.1.2	serialisation	saving/loading checkpoints and passing objects to worker processes	1, 2, 3
TensorBoard (optional)	≥ 2.18	training- curve viewer	only if installed; training runs fine without it	1, 2
The brains	—	trained models	<code>mario_brain_checkpoint.zip</code> (6M, Objective 1) and <code>glitch_hunter_main_brain.zip</code> (16M, Objectives 2–3), frozen and hash-checked	1, 2, 3

Numerical computing, data and storage

Technology	Version	What it is	Where it is used	Objectives
NumPy	2.5.2	arrays and maths	observations, the coverage bitmap, the testable mask, jump arcs, evaluation statistics	1, 2, 3
.npz files	NumPy	compressed array files	coverage state, reachability mask, jump arcs, anchor states (<code>exploration_data/</code>), loaded with pickle disabled	2
Shared memory	stdlib <code>multiprocessing.shared_memory</code>	memory shared between processes	one global coverage map written by all 8 training workers	2
JSON	stdlib	structured records	incident records, manifests, evaluation results, <code>artifacts.json</code> , <code>INJECTED_BUGS.json</code> , <code>level1_design.json</code>	1, 2, 3
JSON Lines (<code>.jsonl</code>)	stdlib	append-only logs	coverage audit trail, reward telemetry, incident occurrences, capture failures	2, 3
SHA-256 (hashlib)	stdlib	file fingerprints	<code>artifacts.json</code> protects the brains and masks; incidents and game variants are identified by hash	2, 3
ZIP	stdlib <code>zipfile</code>	archives	model checkpoints (SB3 format), incident context frames, the "Download all" evidence bundle	1, 2, 3

Exploration and QA logic (the project's own components)

Built in-house on top of the libraries above.

Component	Location	What it does	Objectives
Game environment	<code>custom_mario_env.py</code>	drives the game one frame at a time, reports Mario's position, speed, deaths and glitches	1, 2, 3

Component	Location	What it does	Objectives
Reward wrapper	<code>agent_logic.py</code> , <code>rewards/legacy.py</code> , <code>rewards/qa.py</code>	completion reward (Objective 1); coverage + phase-gated QA reward (Objective 2)	1, 2
Spatial coverage	<code>exploration/coverage.py</code>	1-px world bitmap of everywhere Mario has been, persistent across episodes	2
Reachability / testable mask	<code>exploration/reachability.py</code>	which pixels Mario can physically reach (3 methods + real jump arcs) — the coverage denominator	2
Episode lifecycle	<code>exploration/lifecycle.py</code>	EXPLORE → COMPLETE only on evidence; safety reset on real stagnation, never on time	2
Completion evaluation	<code>evaluation/completion.py</code>	the frozen 500-episode protocol that decides HEALTHY / WARNING / REGRESSED	1, 2
Game variants	<code>reporting/variants.py</code>	identifies clean vs bugged game by tree hash; every bug change must be declared and marked	3
Collision and jump-physics detectors	<code>reporting/collision_invariants.py</code>	generic rules: clipping into solids, blocked by nothing, hit without contact, impossible jump	3
Level design reference	<code>reporting/level_design.py</code> , <code>level1_design.json</code>	the drawn ground, pipes and steps the detectors compare against	3

Image and video processing

Technology	Version	What it is	Where it is used	Objectives
OpenCV (opencv-python)	5.0.0.93	image processing	resizes each game frame to 480×360 and JPEG-encodes it for the live stream; PNG-encodes the exact trigger frame of a bug	1, 3
Pillow	12.3.0	image library	builds the context GIF of each incident and the finish GIF of each clean run, and prepares images for the PDF report	3
Pygame surfaces	2.6.1	raw frame capture	the full-resolution frame copied at the exact moment a detector fires	3

Bug evidence and reporting

Technology / component	Version	What it is	Where it is used	Objectives
fpdf2	2.8.8	PDF generator	the PDF bug report of every incident (<code>reporting/render.py</code>) and the run report of every clean run that reaches the castle (<code>reporting/run_report.py</code>)	3
Markdown	—	text report format	the Markdown bug report of every incident, and of every clean-run report	3
Incident pipeline	<code>reporting/pipeline.py</code>	capture → store → render → replay, in the background	turns each detection into a complete incident; retries and recovers after a crash	3
Incident store	<code>reporting/store.py</code>	evidence on disk	one read-only folder per incident with a hash manifest; atomic writes; never overwritten	3
Clean-run report	<code>reporting/run_report.py</code>	the "No Bugs Found" report	written when a clean-game run reaches the castle: run record, final frame, GIF, Markdown, PDF, downloadable as one zip	3
Incident schema	<code>reporting/schema.py</code>	the standard bug record	ids, timestamps, deterministic JSON, validation	3
Severity, confidence, duplicates	<code>reporting/analysis.py</code>	classification	severity per bug type, detector confidence, same-bug matching and occurrence counts	3
Replay (reproduction)	<code>reporting/reproduce.py</code> + subprocess	re-runs the episode headless	proves the bug happens again on the same frame: reproduced / not reproduced	3
Provenance	<code>reporting/provenance.py</code>	identity of brain, game and code	every report says which brain and which game produced it	3
Maintenance tool	<code>tools/incidents.py</code>	command line	list, show, verify, re-render, re-run replay	3

Web dashboard: backend

Technology / component	Version	What it is	Where it is used	Objectives
Flask	3.1.3	web framework	<code>app.py</code> : serves the page and the REST API (<code>/api/status</code> , <code>/api/incidents</code> , report and evidence downloads)	1, 2, 3
Flask-SocketIO	5.6.1	real-time messaging	pushes video frames, log lines, "Bug Found", run-finished and report-ready events to the browser	1, 2, 3
Threading mode	Flask-SocketIO	concurrency model	real OS threads (eventlet deliberately not used)	1, 2, 3
Werkzeug server	via Flask	local HTTP server	listens on 127.0.0.1:5000 only by default; no debug console	1, 2, 3
Game thread service	<code>dashboard_service.py</code>	one thread owns the game window	Start / Stop / Reset / game switch, pause on every bug, stop at the end of a clean-game run, resume on Start; a page refresh is a Reset	1, 3
Dashboard backend	<code>dashboard_backend.py</code>	env + brain + frame stream	loads the approved brain (hash-checked), plays, captures incidents, writes clean-run reports	1, 2, 3
Safe file serving	<code>reporting/store.py</code> and <code>reporting/run_report.py</code> allow-lists	security	only known incident / run-report ids and file names can be downloaded; path tricks get 404	3

Web dashboard: frontend

Technology	Version	What it is	Where it is used	Objectives
HTML5 (Jinja template)	—	page structure	<code>templates/index.html</code> : controls, video, Bug Tracker, Log Terminal, game selector	1, 2, 3
CSS3	—	styling	<code>static/css/style.css</code> : dark theme and responsive layout	1, 2, 3
Vanilla JavaScript	ES2017+	page behaviour, no framework	<code>static/js/main.js</code> : buttons, live video, Bug Tracker cards, banners (bug found / level complete), reset (also on page refresh), game switch	1, 2, 3

Technology	Version	What it is	Where it is used	Objectives
Socket.IO JS client	vendored (<code>static/</code> <code>vendor/</code>)	browser side of the live connection	receives frames and events; kept locally so the dashboard works offline	1, 2, 3
Fetch API	browser	REST calls	loads status and the session's incidents	3

Testing, code quality and CI/CD

Technology	Version	What it is	Where it is used	Objectives
pytest	9.1.1	test framework	~785 automated tests in <code>tests/</code> (fast ones by default, <code>-m slow</code> for brain and end-to-end)	1, 2, 3
pytest-cov	7.1.0	code coverage	the full gate requires at least 90% (last run 90.53%, measured to two decimals)	2, 3
Ruff	0.16.6	linter	style, bugs and security (bandit) rules on every run	2, 3
mypy	2.3.1	static type checker	strict mode, checked for both Windows and Linux	2, 3
pre-commit	4.6.2	git hooks	runs checks before each commit	2, 3
Quality gate	<code>tools/</code> <code>check.py</code>	one command for all checks	lint → types → tests → artifact hashes (<code>--</code> <code>full</code> adds slow tests + coverage floor)	2, 3
Artifact verification	<code>tools/</code> <code>verify_</code> <code>artifacts.py</code>	integrity check	re-hashes the protected brains, masks and baseline against <code>artifacts.json</code>	2, 3
End-to-end validation	<code>tools/</code> <code>validate_</code> <code>incident_</code> <code>pipeline.py</code>	37-check pipeline proof	runs the real dashboard stack headless and checks every reporting stage	3
Git + GitHub	—	version control and hosting	source history and the remote repository	1, 2, 3
GitHub Actions	<code>.github/</code> <code>workflows/</code> <code>ci.yml</code>	continuous integration	lint, types and tests on Ubuntu and Windows on every push	2, 3

Tooling, scripts and documentation

Tool / format	Where it is used	Objectives
<code>train_agent.py</code>	training, with safe resume (<code>--resume-from</code> , <code>--dry-run-resume</code>)	1, 2
<code>tools/evaluate_completion.py</code>	500-episode completion test (HEALTHY / WARNING / REGRESSED)	1, 2
<code>tools/build_reachability.py</code> , <code>collect_jump_arcs.py</code> , <code>bootstrap_coverage.py</code> , <code>migrate_coverage.py</code> , <code>build_anchor_set.py</code> , <code>calibrate_*.py</code>	one-time setup and measurement for the QA campaign	2
<code>tools/build_level_design.py</code>	writes the level-design reference from the clean game	3
<code>run_dashboard.bat</code>	double-click to start the dashboard and open the browser	3
Markdown docs	<code>README.md</code> , <code>docs/OBJECTIVE1-3.md</code> , <code>docs/ARCHITECTURE.md</code> , <code>docs/PERFORMANCE.md</code> , <code>SECURITY.md</code> , <code>CONTRIBUTING.md</code>	1, 2, 3
Mermaid diagrams	the architecture diagram in <code>docs/ARCHITECTURE.md</code>	2, 3

Project map: every folder and file

Every folder and file in the project, tracked in git or not, with what it holds and what uses it. "In git" says whether the file is committed; gitignored files exist only on the machine that made or installed them. Large generated sets (the Python environment, caches) are described as a folder.

Folders at a glance

Folder	What it holds	Where it is used	In git
<code>common/</code>	small shared helpers: logging, atomic file writes, hashes, tool start-up	every entry point and tool	yes
<code>rewards/</code>	the two reward functions: completion (Objective 1) and QA exploration (Objective 2)	<code>agent_logic.py</code> during training and play	yes
<code>training/</code>	Stable-Baselines3 callbacks, checkpoint helpers, the one-off value-head reset	<code>train_agent.py</code>	yes

Folder	What it holds	Where it is used	In git
evaluation/	the 500-episode completion protocol, Level-1 verification, the frozen 6M baseline	training resume checks, <code>tools/evaluate_completion.py</code> , <code>tools/verify_level1.py</code>	yes (<code>results/</code> is not)
exploration/	<code>config.py</code> (every tuned value), spatial coverage, reachability mask, episode lifecycle, level completion	the environment, reward wrapper, training, dashboard	yes
reporting/	Objective 3: detectors, incident pipeline, store, reports, replay, run reports, game variants	dashboard, tests, <code>tools/incidents.py</code>	yes
templates/, static/	the dashboard web page: HTML, CSS, JavaScript, bundled Socket.IO client	served by <code>app.py</code>	yes
assets/	the README's screenshots and GIF	<code>README.md</code>	yes
docs/	Objective 1-3 summaries, architecture, performance notes	readers; linked from <code>README.md</code>	yes
tools/	command-line tools: quality gate, evaluation, data builders, incident tools, installer	run by hand and by <code>tools/check.py</code> / tests	yes
tests/	about 785 automated tests and their helpers	<code>pytest</code> , <code>tools/check.py</code> , CI	yes
.github/	the CI workflow	GitHub Actions on every push	yes
mario_clean/	the original Mario clone (code, graphics, music, sounds), unmodified and hash-pinned	the game every brain is trained and measured on	yes
mario_bugged/	a copy of the clone with the 6 declared benchmark bugs	the bug-detection target, picked in the dashboard	yes
checkpoints_qa/	the final 16M brain's closure record, evidence and backup, and the 6.03M anchor checkpoint	dashboard approval check, retraining with anchor-KL, <code>artifacts.json</code>	no
exploration_data/	measured data: reachability mask, jump arcs, observed reach, coverage bootstrap, anchor states	QA reward, coverage, anchor-KL training	no
coverage_audits/	coverage maps written by <code>tools/remaining_coverage_map.py</code>	reading only (no code loads them)	no
incidents/	one evidence folder per bug found, plus the repeat-sighting log	dashboard Bug Tracker, <code>tools/incidents.py</code>	no

Folder	What it holds	Where it is used	In git
<code>run_reports/</code>	one "No Bugs Found" report folder per clean run that reached the castle	dashboard banner and Bug Tracker	no
<code>venv_gpu/</code>	the Python environment: Python 3.12 and every installed library (about 4.4 GB)	every command	no
<code>__pycache__/</code> , <code>.mypy_cache/</code> , <code>.ruff_cache/</code> , <code>.pytest_cache/</code>	tool caches, rebuilt automatically	Python, mypy, ruff, pytest	no
<code>.git/</code>	the repository's full history	git	(git's own folder)

Files in the project root

File	What it is and what it holds	Where it is used	In git
<code>app.py</code>	the dashboard server (Flask + Socket.IO): page, REST API, report and evidence downloads, socket events	started by <code>python app.py</code> or <code>run_dashboard.bat</code>	yes
<code>dashboard_service.py</code>	the single game thread: Start / Stop / Reset / game switch, pause on bugs, stop at a clean run's end, step pacing	<code>app.py</code>	yes
<code>dashboard_backend.py</code>	loads the approved brain, runs the game, streams frames, captures incidents, writes run reports	<code>dashboard_service.py</code> , <code>app.py</code>	yes
<code>desktop.py</code>	<code>run_dashboard.bat</code> 's helpers: full speed on battery (power mode, no throttling), centred terminal and browser	<code>run_dashboard.bat</code> , <code>app.py --desktop</code>	yes
<code>game_window.py</code>	Windows window calls for the game window: centre, show, hide, minimise to the taskbar	<code>custom_mario_env.py</code> , <code>desktop.py</code>	yes
<code>custom_mario_env.py</code>	the game as a Gymnasium environment: actions, observations, info, built-in detectors, the game window's life cycle	training, dashboard, evaluation, tests, tools	yes
<code>agent_logic.py</code>	<code>GlitchHunterWrapper</code> : coverage recording, the episode lifecycle and the reward	training, dashboard, evaluation	yes

File	What it is and what it holds	Where it is used	In git
<code>train_agent.py</code>	training for Objectives 1 and 2, with safe resume	run by hand (<code>python train_agent.py</code>)	yes
<code>run_dashboard.bat</code>	Windows double-click launcher for the dashboard	the user	yes
<code>mario_brain_checkpoint.zip</code>	the 6M brain (Objective 1), 20 MB: the fallback brain	dashboard when the 16M brain is absent; baseline for evaluation	yes (the one large file tracked)
<code>glitch_hunter_main_brain.zip</code>	the final 16M brain (Objectives 2-3), 20 MB, hash-checked	dashboard's default brain; a resume candidate for training	no (installed by <code>tools/final_brain.py</code>)
<code>glitch_hunter_main_brain_coverage.npz</code>	the 16M brain's coverage map (where it has been in the level)	dashboard's QA reward, coverage maps	no
<code>artifacts.json</code>	SHA-256 of every protected brain, mask, baseline and evidence file	<code>tools/verify_artifacts.py</code> , <code>tools/final_brain.py</code> , tests	yes
<code>pyproject.toml</code>	dependencies and the settings for ruff, mypy, pytest and coverage	<code>uv sync</code> , every quality tool, CI	yes
<code>uv.lock</code>	exact locked versions of every dependency	<code>uv sync --active</code>	yes
<code>requirements.txt</code>	the same dependencies for plain pip, with the install steps	<code>pip install -r requirements.txt</code> , CI	yes
<code>README.md</code>	the main guide: overview, installation, how to run, results	readers, GitHub front page	yes
<code>CONTRIBUTING.md</code>	how to develop: set-up, quality gate, code rules, commits, releasing the brain	contributors	yes
<code>SECURITY.md</code>	what the dashboard exposes, what <code>run_dashboard.bat</code> changes, dependency audit	readers	yes
<code>THIRD_PARTY_NOTICES.md</code>	terms of the Mario clone, Nintendo assets and every library	readers	yes
<code>LICENSE</code>	MIT licence for the project's own work, and what it does not cover	readers	yes
<code>Glitch_Hunter_Tech_Stack.pdf</code>	this document as a PDF	readers	yes

File	What it is and what it holds	Where it is used	In git
<code>Glitch_Hunter_Panel_QA.pdf</code>	150 questions a presentation panel may ask about the project, with simple-English answers (purpose, AI, the three objectives, the START TESTING code path, making it generic, related work, limits)	presenters, before a viva or review	no (untracked)
<code>.gitignore</code>	what git must never commit, and why	git	yes
<code>.gitattributes</code>	line-ending and binary rules	git	yes
<code>.pre-commit-config.yaml</code>	optional checks run on every commit (ruff, mypy, file hygiene)	<code>pre-commit</code>	yes
<code>.coverage</code>	test-coverage data from the last full gate	<code>tools/check.py --full</code> ; rebuilt each run	no
<code>.dashboard_power.json</code>	your original Windows power modes, kept only until they are put back	<code>desktop.py</code> ; exists only briefly	no

Core source packages

All tracked in git; each folder's `__init__.py` marks it as a Python package and says what it is for.

File	What it is and what it holds	Where it is used
<code>common/cli.py</code>	the start-up every tool shares: project root on the path, headless SDL, logging	every script in <code>tools/</code>
<code>common/fileio.py</code>	atomic file writes, SHA-256 hashes, read-only protection for artifacts	training, reporting, tools
<code>common/logging_setup.py</code>	one logging configuration and the progress log	every entry point
<code>rewards/legacy.py</code>	the completion reward, unchanged from the 6M training run	<code>agent_logic.py</code> in completion mode
<code>rewards/qa.py</code>	the QA exploration reward: new coverage, phase-gated completion bonus	<code>agent_logic.py</code> in QA mode
<code>rewards/shared.py</code>	state and shaping both rewards share, incl. the adaptive death memory	both reward modes
<code>training/callbacks.py</code>	the training callbacks: checkpoints with coverage, anchor-KL, stagnation watch, telemetry	<code>train_agent.py</code>
<code>training/checkpoints.py</code>	reads a checkpoint's step count without loading it; retention verdicts for resume	<code>train_agent.py</code>

File	What it is and what it holds	Where it is used
<code>training/value_head.py</code>	the one-off critic reset when QA training starts from the 6M brain	<code>train_agent.py</code>
<code>evaluation/completion.py</code>	the frozen 500-episode test: can a brain still finish Level 1-1?	<code>tools/evaluate_completion.py</code> , resume checks
<code>evaluation/level1_verification.py</code>	decides whether a Level-1 snapshot is the final brain	<code>tools/verify_level1.py</code>
<code>evaluation/completion_baseline_6M.json</code>	the 6M brain's frozen completion result (46.8%) that others are compared with	<code>evaluation/completion.py</code> ; hash-protected
<code>exploration/config.py</code>	every tuned value (rewards, thresholds, paths), each with the measurement behind it	nearly every module
<code>exploration/coverage.py</code>	the world-space coverage bitmap, shared by all training workers	reward wrapper, training, dashboard, tools
<code>exploration/reachability.py</code>	the testable-pixel mask: which pixels Mario can physically reach	coverage percentages, <code>tools/build_reachability.py</code>
<code>exploration/lifecycle.py</code>	the episode lifecycle: EXPLORE to COMPLETE, and how an episode ends	reward wrapper, dashboard
<code>exploration/level_completion.py</code>	when Level 1 counts as fully covered; coverage maps and region lists	training, <code>tools/remaining_coverage_map.py</code>

`evaluation/results/` (not in git) holds one completion-test result per evaluated brain; the 16M brain's result is hash-protected and read when choosing what to resume.

reporting/ (Objective 3)

All tracked in git.

File	What it is and what it holds	Where it is used
<code>__init__.py</code>	package marker: turns a detected anomaly into a reviewable incident	-
<code>events.py</code>	the <code>Detection</code> a detector produces, and the synthetic test probe	environment, pipeline
<code>collision_invariants.py</code>	the generic bug detectors: clipping into solids, blocked by nothing, hit without contact, impossible jump	<code>custom_mario_env.py</code>
<code>level_design.py</code>	loads Level 1-1 as designed, for the detectors to compare against	<code>collision_invariants.py</code>
<code>level1_design.json</code>	the drawn ground, pipes and steps (37 solids) measured from the clean game	<code>level_design.py</code> ; rebuilt only by <code>tools/build_level_design.py</code>

File	What it is and what it holds	Where it is used
<code>pipeline.py</code>	capture, then GIF, reports and replay in the background; duplicate counting; session view	dashboard backend
<code>evidence.py</code>	writes the raw evidence: trigger frame, context frames, trajectory, the incident record	<code>pipeline.py</code>
<code>schema.py</code>	the incident record's format, ids and validation	pipeline, store, tools
<code>store.py</code>	the incident folders on disk: atomic, read-only, never overwritten; safe file lookup	pipeline, <code>app.py</code> , <code>tools/incidents.py</code>
<code>analysis.py</code>	severity, detector confidence and the duplicate fingerprint	<code>pipeline.py</code>
<code>render.py</code>	an incident's GIF, Markdown report and PDF report	<code>pipeline.py</code> , <code>tools/incidents.py</code>
<code>reproduce.py</code>	replays an incident's episode in a separate process and says whether it happened again	<code>pipeline.py</code>
<code>provenance.py</code>	which brain, game and code produced an incident	dashboard backend, pipeline
<code>variants.py</code>	the clean and bugged game variants, their tree hashes and declared bugs	environment, provenance, tests
<code>run_report.py</code>	the clean game's "No Bugs Found" run report: record, final frame, GIF, Markdown, PDF	dashboard backend, <code>app.py</code>

Dashboard page, images, docs and CI

All tracked in git.

File	What it is and what it holds	Where it is used
<code>templates/index.html</code>	the dashboard page: controls, video, Bug Tracker, Log Terminal, game selector, banners	served by <code>app.py</code> at <code>/</code>
<code>static/js/main.js</code>	the page's behaviour: buttons, live video, Bug Tracker cards, banners, reset (also on refresh), game switch	loaded by <code>index.html</code>
<code>static/css/style.css</code>	the page's dark theme and layout	loaded by <code>index.html</code>
<code>static/vendor/socket.io.min.js</code>	the Socket.IO browser client (v4.7.2), bundled so the page works offline	loaded by <code>index.html</code>
<code>assets/dashboard.png</code>	screenshot of the dashboard	<code>README.md</code>

File	What it is and what it holds	Where it is used
<code>assets/bug-found.png</code>	screenshot of a "Bug Found" stop	<code>README.md</code>
<code>assets/bug-stair-clip.gif</code>	recording of the stair-clipping bug being caught	<code>README.md</code>
<code>docs/OBJECTIVE1.md</code>	Objective 1: the 6M brain, how it was trained, its results	readers; <code>README.md</code>
<code>docs/OBJECTIVE2.md</code>	Objective 2: QA exploration, audits, the final 16M brain	readers; <code>README.md</code>
<code>docs/OBJECTIVE3.md</code>	Objective 3: detectors, reporting pipeline, benchmark bugs, run reports	readers; <code>README.md</code>
<code>docs/ARCHITECTURE.md</code>	components, invariants, protected artifacts, where to change what	readers; <code>README.md</code>
<code>docs/PERFORMANCE.md</code>	time and memory per training worker, and how to measure them	readers; <code>README.md</code>
<code>.github/workflows/ci.yml</code>	continuous integration: lint, types and tests on Ubuntu and Windows	GitHub Actions on every push

tools/ (command-line tools)

All tracked in git; each is run by hand as `python tools/<name>.py`, and `tests/test_tools_cli.py` checks every one starts and documents itself.

File	What it does	What it writes
<code>README.md</code>	the guide to every tool below	-
<code>check.py</code>	every quality gate CI runs: ruff, mypy, tests, artifact hashes (<code>--full</code> : slow tests + coverage floor)	nothing
<code>final_brain.py</code>	installs the final 16M brain from the GitHub Release (verified against <code>artifacts.json</code>), or builds that release zip	the four final-brain files
<code>verify_artifacts.py</code>	re-hashes every protected artifact against <code>artifacts.json</code>	nothing (<code>--record</code> rewrites the manifest)
<code>benchmark_step.py</code>	milliseconds per agent step, and where the time goes	nothing
<code>evaluate_completion.py</code>	can a brain still finish Level 1-1? HEALTHY / WARNING / REGRESSED against the 6M baseline	<code>evaluation/results/</code>
<code>verify_level1.py</code>	checks a Level-1 snapshot before it counts as the final brain	a verification JSON beside the snapshot
<code>remaining_coverage_map.py</code>	where the uncovered reachable pixels are, as a picture and a region list	<code>coverage_audits/</code>

File	What it does	What it writes
<code>incidents.py</code>	the incident store: list, show, verify, re-render, reproduce, recover	only new versioned files, never overwrites
<code>validate_incident_pipeline.py</code>	37-check end-to-end proof of the reporting pipeline on the real dashboard stack	its own store under <code>incidents/_validation/</code>
<code>build_level_design.py</code>	writes Level 1-1's drawn solids from the clean game	<code>reporting/level1_design.json</code>
<code>collect_jump_arcs.py</code>	records the engine's real jump arcs	<code>exploration_data/jump_arcs.npz</code>
<code>build_reachability.py</code>	builds the reachable-pixel mask (the coverage denominator)	<code>exploration_data/reachable_mask.npz</code>
<code>build_anchor_set.py</code>	records the anchor states a healthy brain sees, for anchor-KL training	<code>exploration_data/anchor_states.npz</code>
<code>migrate_coverage.py</code>	re-stamps a coverage file to a rebuilt mask, losslessly	a new coverage file
<code>bootstrap_coverage.py</code>	seeds the coverage map with the 6M brain's usual routes	<code>exploration_data/coverage_bootstrap*.npz</code>
<code>calibrate_reward.py</code>	solves the QA reward's scale against the completion reward	a value in <code>exploration/config.py</code>
<code>calibrate_phase_reward.py</code>	measures every QA reward channel on real runs (source of the phase-gated values)	<code>calibration_runs/</code>
<code>objective2_evidence/jump_physics.py</code>	engine ground truth for the 172 px obstacles (tall pipes, staircase top)	nothing
<code>objective2_evidence/runup.py</code>	the minimum run-up for a full-height jump	nothing
<code>objective2_evidence/well_escape.py</code>	exhaustive search: is the staircase well a soft-lock?	nothing
<code>objective2_evidence/well_trace.py</code>	replays one well-escape attempt frame by frame	nothing

tests/

All tracked in git and run by `pytest` (through `tools/check.py` and CI). The first six are helpers, not tests.

File	What it holds or checks
<code>conftest.py</code>	shared fixtures (the real game environment, temporary folders) for every test

File	What it holds or checks
<code>incident_helpers.py</code>	builds realistic detections and capture context without running the game; used by the incident tests
<code>mp_workers.py</code>	worker processes for the multi-process coverage test
<code>bug_probe.py</code>	plays every injected-bug scenario on one game variant; run by <code>test_injected_bugs.py</code>
<code>variant_probe.py</code>	plays one game variant and prints its fingerprint; run by <code>test_game_variants.py</code>
<code>bug_incident_run.py</code>	one dashboard episode through the real reporting pipeline; run by <code>test_bug_incidents.py</code>
<code>test_arc_reach.py</code>	reachability from measured jump arcs instead of a rectangle
<code>test_artifacts.py</code>	protected files are still the recorded bytes; the brain backups match
<code>test_bug_incidents.py</code>	all six benchmark bugs become reproduced, reported incidents; the clean game reports none
<code>test_collision_invariants.py</code>	the collision and jump detectors, with a regression test per false alarm
<code>test_common.py</code>	<code>common/</code> : atomic file I/O, logging, tool start-up
<code>test_completion_eval.py</code>	the 500-episode completion evaluation
<code>test_concurrency.py</code>	thread-safety of the dashboard server
<code>test_coverage.py</code>	coverage recording, saving, frontier, adaptive target
<code>test_coverage_env.py</code>	coverage from the real game is in world space
<code>test_coverage_multiproc.py</code>	coverage is shared across real OS processes
<code>test_dashboard_backend.py</code>	what the browser is sent, clean-run endings and reports, refresh = reset, <code>--desktop</code> mode
<code>test_dashboard_control.py</code>	the game thread and window: Start, Stop, X, Reset, game switch, pacing, minimised start
<code>test_dashboard_incidents.py</code>	the dashboard side of Objective 3: stops, Bug Tracker, downloads
<code>test_desktop.py</code>	<code>desktop.py</code> : power-mode guard, restorer, centred windows
<code>test_env.py</code>	the environment's contract and past-bug regressions
<code>test_episode_lifecycle.py</code>	the engine timer, EXPLORE to COMPLETE, and how episodes end
<code>test_final_brain.py</code>	<code>tools/final_brain.py</code> installs the 16M brain intact
<code>test_game_variants.py</code>	the clean game is untouched; the bugged game differs only where declared
<code>test_game_window.py</code>	show, hide and minimise calls of the game window

File	What it holds or checks
<code>test_glitch_detection.py</code>	the built-in detectors behind the Bug Tracker
<code>test_hud_timer.py</code>	the on-screen TIME box during long QA episodes
<code>test_incident_capture.py</code>	capturing evidence from the real game
<code>test_incident_pipeline.py</code>	schema, store, duplicates, reports, without the game
<code>test_incident_reproduce.py</code>	replay is honest both ways (reproduced / not reproduced)
<code>test_incident_tools.py</code>	<code>tools/incidents.py</code> and the end-to-end validation
<code>test_injected_bugs.py</code>	each of the six bugs, proved against the clean game
<code>test_level1_compliance.py</code>	what surrounds Level-1 completion
<code>test_level_completion.py</code>	QA training ends on Level-1 coverage, not a step count
<code>test_migrate_coverage.py</code>	<code>tools/migrate_coverage.py</code>
<code>test_noncoverage_taxonomy.py</code>	the two false-coverage causes found during Objective 2
<code>test_objective2_protection.py</code>	Objective 3 never writes into Objective 2's files
<code>test_phase_reward.py</code>	the phase-gated QA reward
<code>test_qa_resume.py</code>	QA checkpoints, resuming, and the 6M brain's safety
<code>test_quality_gate.py</code>	the coverage floor is measured to two decimals
<code>test_reward_calibration.py</code>	the calibrated reward and its bookkeeping
<code>test_reward_qa.py</code>	reward shaping in QA mode
<code>test_reward_telemetry.py</code>	the reward telemetry written during training
<code>test_reward_wrapper.py</code>	reward shaping in completion (legacy) mode
<code>test_rewards_death_memory.py</code>	the adaptive death memory
<code>test_run_report.py</code>	the "No Bugs Found" run report, its files and downloads
<code>test_testable_mask.py</code>	the coverage denominator never exceeds the real world
<code>test_tools_cli.py</code>	every tool starts, parses its options and documents itself
<code>test_train_agent_stages.py</code>	<code>train_agent.py</code> : resume, device, callbacks, banner
<code>test_training_callbacks.py</code>	the training callbacks and checkpoint helpers

The game: mario_clean/ and mario_bugged/

Both folders hold the same third-party Mario clone and are tracked in git. `mario_clean/` is never edited (its whole tree is pinned by SHA-256); `mario_bugged/` differs only where `INJECTED_BUGS.json` declares a bug. The game code is loaded by `custom_mario_env.py`.

File (in each folder)	What it is and what it holds
<code>VARIANT.md</code>	what this copy is for; in <code>mario_bugged/</code> , the six bugs and the rules for changing it
<code>INJECTED_BUGS.json</code> (<code>mario_bugged/</code> only)	the declaration of every benchmark bug: where, which lines, the pinned diff hash
<code>data/__init__.py</code> , <code>data/components/__init__.py</code> , <code>data/states/__init__.py</code>	package markers
<code>data/setup.py</code>	starts pygame, opens the game window, loads every image, song and sound
<code>data/tools.py</code>	<code>Control</code> , the game's main loop and state machine; the image/music/sound loaders
<code>data/constants.py</code>	screen size, colours, physics values, state names
<code>data/game_sound.py</code>	which music and sound effects play when
<code>data/states/level1.py</code>	Level 1-1 itself: builds ground, pipes, steps, bricks and enemies, and runs every frame's collisions (in <code>mario_bugged/</code> , carries five of the six bugs: stair, pipe and ceiling clipping, invisible wall, false Goomba hit)
<code>data/components/mario.py</code>	Mario: movement, jumping, growing, dying, animation (in <code>mario_bugged/</code> , carries the open-sky jump bug)
<code>data/components/enemies.py</code>	Goombas and Koopas
<code>data/components/bricks.py</code>	bricks and the pieces they break into
<code>data/components/coin_box.py</code>	? boxes
<code>data/components/coin.py</code> , <code>data/components/flashing_coin.py</code>	coins that pop out of boxes; the blinking coin in the score bar
<code>data/components/powerups.py</code>	mushroom, 1-up, fire flower, star, fireball
<code>data/components/collider.py</code>	invisible solid rectangles: ground, pipes, steps
<code>data/components/checkpoint.py</code>	invisible triggers that release enemies and mark events
<code>data/components/flagpole.py</code> , <code>data/components/castle_flag.py</code>	the flagpole at the end and the castle flag
<code>data/components/info.py</code>	the score bar: score, coins, world, time

File (in each folder)	What it is and what it holds
<code>data/components/score.py</code>	the floating score numbers
<code>resources/graphics/level_1.png</code>	the level background
<code>resources/graphics/mario_bros.png</code>	Mario's sprites
<code>resources/graphics/smb_enemies_sheet.png</code>	enemy sprites
<code>resources/graphics/tile_set.png</code>	bricks, boxes and tiles
<code>resources/graphics/item_objects.png</code>	coins, power-ups, flag
<code>resources/graphics/text_images.png</code>	the score bar's font
<code>resources/music/</code>	8 tracks: <code>main_theme</code> , <code>main_theme_spd_up</code> , <code>invincible</code> , <code>game_over</code> (.ogg); <code>death</code> , <code>flagpole</code> , <code>out_of_time</code> , <code>stage_clear</code> (.wav)
<code>resources/sound/</code>	13 effects (.ogg): <code>big_jump</code> , <code>small_jump</code> , <code>brick_smash</code> , <code>bump</code> , <code>coin</code> , <code>count_down</code> , <code>fireball</code> , <code>kick</code> , <code>one_up</code> , <code>pipe</code> , <code>powerup</code> , <code>powerup_appears</code> , <code>stomp</code>

The images and sounds are Nintendo's (see `THIRD_PARTY_NOTICES.md`). Each folder's `__pycache__` (not in git) holds compiled game code.

Brains, measured data and training evidence (not in git)

Too large for git; the important ones are hash-pinned in `artifacts.json` and checked by `tools/verify_artifacts.py`. The final brain's files are installed by `tools/final_brain.py`.

File	What it is and what it holds	Where it is used
<code>checkpoints_qa/final_objective2_16000000/FINAL_OBJECTIVE2.json</code>	Objective 2's closure record: the approved 16M brain's hash and results	the dashboard approves the brain against it
<code>.../final_objective2_16000000/glitch_hunter_main_brain.zip</code> , <code>..._coverage.npz</code>	a protected backup of the final brain and its coverage	restore source; <code>tests/test_artifacts.py</code>
<code>.../evidence/train.log</code>	the final training run's log	evidence of how the 16M brain was made
<code>.../evidence/console.log</code>	the final training run's console output	evidence
<code>.../evidence/reward_telemetry.jsonl</code>	every reward channel, per episode, for the whole run (17 MB)	evidence; reward analysis

File	What it is and what it holds	Where it is used
.../evidence/coverage_audit_trail.jsonl	coverage at every checkpoint of the run	evidence; coverage history
.../evidence/glitch_hunter_qa_16000000_steps_d906d09e.json	the 16M brain's 500-episode completion result (56.6%)	evidence
.../evidence/completion_baseline_6M.json	the 6M baseline as it stood at closure	evidence
checkpoints_qa/pre_main_6032768/glitch_hunter_qa.zip, ..._coverage.npz	the 6.03M "healthy" checkpoint: the reference brain for anchor-KL	loaded by anchor-KL training (the final run used it)
checkpoints_qa/pre_main_6032768/reward_telemetry.jsonl, coverage_audit_trail.jsonl	that checkpoint's own telemetry and coverage trail	record only
exploration_data/reachable_mask.npz	the reachable-pixel mask: the denominator of every coverage %	coverage, QA reward, tools
exploration_data/jump_arcs.npz	the engine's real jump arcs (600 take-offs)	building the mask
exploration_data/observed_reach.npz	places real play reached that the arc model had to keep	building the mask
exploration_data/coverage_bootstrap_6000000_mask_v4.npz	the 6M brain's routes, seeding the QA coverage map	QA training start
exploration_data/anchor_states.npz	6,007 frames the healthy brain sees, where anchor-KL is measured	anchor-KL training
evaluation/results/glitch_hunter_qa_16000000_steps_d906d09e.json	the 16M brain's completion verdict	read when choosing what to resume
coverage_audits/glitch_hunter_main_brain_coverage_remaining_map.png	the 16M brain's coverage map: red = reachable but never visited (84.25% covered)	reading only
coverage_audits/glitch_hunter_main_brain_coverage_remaining_regions.json	the uncovered regions of that map, with sizes and positions	reading only
coverage_audits/arc_reach_map_14.8M.png	an older coverage map from the 14.8M training point	reading only (superseded)

Runtime output: incidents/ and run_reports/ (not in git)

Written by the dashboard while it tests; kept on this machine and never overwritten. Right now `incidents/` holds 8 real bugs found on the bugged game (all six bug types) and `run_reports/` holds 4 clean runs.

File	What it is and what it holds	Where it is used
<code>incidents/INC-<date>-<time>-<id>/</code>	one folder per bug found, read-only once written	Bug Tracker, <code>/incidents/...</code> downloads, <code>tools/incidents.py</code>
<code>.../incident.json</code>	the full record: detector, measurements, position, game and brain identity	reports, Bug Tracker
<code>.../trigger.png</code>	the exact frame the bug was detected, full resolution	reports, "Trigger frame" link
<code>.../context_frames.zip</code>	the stream frames just before the trigger	the GIF, the PDF film strip
<code>.../trajectory.json</code>	Mario's state and the action taken, frame by frame, before the trigger	the reports; replay
<code>.../context.gif</code>	the moments before the bug, then the trigger frame held	"GIF" link
<code>.../report.md</code> , <code>.../report.pdf</code>	the bug report in Markdown and PDF	"Markdown / PDF report" links
<code>.../reproduction.json</code>	the replay's verdict: reproduced or not	reports, Bug Tracker
<code>.../manifest.json</code>	SHA-256 of every file, and how each report was produced	<code>tools/incidents.py</code> <code>verify</code>
<code>incidents/occurrences.jsonl</code>	every later sighting of an already-known bug (raises its "Seen" count)	Bug Tracker counts
<code>incidents/_validation/</code>	appears only after <code>tools/validate_incident_pipeline.py</code> ; its own test bugs, hidden from the dashboard	the validation tool
<code>run_reports/RUN-<date>-<time>-<id>/</code>	one folder per clean-game run that reached the castle	Level Complete banner, Bug Tracker
<code>.../run.json</code>	the run's record: outcome, length, furthest point, score, bugs (none)	the reports
<code>.../final.png</code>	the last frame of the run, full resolution	"Final frame" link
<code>.../finish.gif</code>	the run's last seconds	"GIF" link
<code>.../report.md</code> , <code>.../report.pdf</code>	the "No Bugs Found" report in Markdown and PDF	report links, "Download all (.zip)"

Environment and caches (not in git)

All rebuilt automatically or by the install steps; described as folders because they hold thousands of generated files.

Folder or file	What it is and what it holds	Where it is used
<code>venv_gpu/</code>	the project's Python 3.12.10 environment, about 26,500 files and 4.4 GB	every command, <code>run_dashboard.bat</code>
<code>venv_gpu/Lib/site-packages/</code>	the 130 installed packages: PyTorch 2.14.0 + CUDA 12.6 (4.0 GB of it), Stable-Baselines3, pygame, Flask, NumPy, OpenCV, fpdf2, pytest, ruff, mypy and their dependencies	imported by all project code
<code>venv_gpu/Scripts/</code>	<code>python.exe</code> , <code>pythonw.exe</code> (no console, used for background helpers), <code>pip</code> , <code>pytest</code> , <code>ruff</code> , <code>mypy</code> , <code>pre-commit</code> and activation scripts	every command
<code>venv_gpu/pyvenv.cfg</code> , <code>Include/</code> , <code>share/</code>	the environment's settings (which base Python) and package extras	Python itself
<code>__pycache__/</code> (root and every code folder)	compiled <code>.pyc</code> copies of the Python files	Python, to start faster
<code>.mypy_cache/</code>	mypy's type-check cache	<code>mypy</code> , <code>tools/check.py</code>
<code>.ruff_cache/</code>	ruff's lint cache	<code>ruff</code>
<code>.pytest_cache/</code>	pytest's cache (last failures); appears when pytest runs without <code>-p no:cacheprovider</code>	<code>pytest</code>
<code>.git/</code>	the full version history and GitHub link (about 80 MB)	git

Notes

- **torch is pinned to 2.14.0** (CUDA 12.6 build): it has no published security advisories, and it meets Stable-Baselines3's own torch requirement, so SB3 installs normally. It replaced 2.5.1 (CUDA 12.1) on 2026-09-25; both brains play move-for-move the same on it. Install torch first (so Windows gets the GPU build), then `requirements.txt` — or simply `uv sync --active`.
- **eventlet must never be installed.** Flask-SocketIO picks its async mode from what is installed; the dashboard runs in plain threading mode on purpose.
- **Headless everywhere except the dashboard:** training, evaluation, tests and incident replays use the SDL dummy video driver; only the dashboard opens a real game window.
- **Artifacts are protected by hash, not by git:** the brains, masks and coverage files are large and git-ignored; `artifacts.json` (tracked) holds their SHA-256 values.